Gauge vacuum estimates: global tails and conditional slice transfer

On a fixed finite Kogut–Susskind lattice, the weighted eigenfunction identity gives an integrated tail for **total magnetic-energy excess**:

$$\mathbb{P}_\Omega(V - e_0/B' \geq s) \leq \min\left\{1, C_{\varepsilon_E,\delta} e^{-8(1-\delta)s^{3/2}/(3g^2M)}\right\}, \qquad M \geq \|\nabla V\|_\infty > 0.$$

All constants and volume dependence are given below. The estimate is neither pointwise wavefunction decay nor a volume-uniform probability bound for a local excitation. Exact magnetic identities and a second exponential comparison make the same global-data limitation explicit. They do not prove that every Hamiltonian approach to local control fails.

A different exact statement identifies the Perron-vector density of a fixed-spacing positive transfer matrix with its infinite-time slice marginal. The analogous Hamiltonian-semigroup identity holds directly. Connecting the anisotropic Wilson family to that Hamiltonian still requires projector convergence; importing a local estimate requires the same observable and uniform constants in the claimed limits. None of these identities supplies an interacting rarity bound, signed operator comparison or physical continuum gap.

Common setting and magnetic identities

Fix $G = SU(N)$, $N \geq 2$, $g, a > 0$, and a finite periodic three-dimensional spatial cubic lattice with at least three sites per axis, so elementary plaquettes have four distinct links. Put $\mathcal{M} = G^{\mathcal{E}}$ with normalized product Haar measure and the product bi-invariant metric, $\mathrm{tr}(T^a T^b) = \delta^{ab}/2$. With $-\Delta \geq 0$,

$$H_{\mathrm{KS}} = \frac{\hbar c}{a}\{A'(-\Delta) + B'V\}, \quad A' = g^2/2, \quad B' = 2/g^2, \quad V = \sum_p (N - \mathrm{Re}\,\mathrm{tr}\, U_p).$$

The finite-lattice theorem gives a smooth positive normalized gauge-invariant ground state Ω, with $e_0 = aE_0/(\hbar c)$ and $-A'\Delta\Omega + (B'V - e_0)\Omega = 0$. In particular

$$\bar{v} := e_0/B' = \langle V\rangle_\Omega + (A'/B')\int |\nabla\Omega|^2 d\mu \geq \langle V\rangle_\Omega.$$

The threshold is not the mean magnetic energy. The Agmon window $\varepsilon_E > 0$ below is an energy threshold in lattice units, unrelated to the classical parameter $\epsilon = \hbar/S_0$.

The Laplacian identity

Laplacian identity. On $\mathcal{M} = G^{\mathcal{E}}$ with the product bi-invariant metric,

$$\Delta V = 4C_2(R_{\mathrm{f}})(N|\mathcal{P}| - V),$$

where $C_2(R_{\mathrm{f}})$ is the quadratic Casimir of the defining representation. Equivalently $W = V - N|\mathcal{P}|$ satisfies $\Delta W = -4C_2 W$: the magnetic energy, shifted by its Haar mean, is an eigenfunction of the Laplacian.

Proof. Fix a plaquette p and one of its four links ℓ. Writing $\mathrm{tr}\, U_p = \sum_{ab}(U_\ell)_{ab} M_{ba}$ with M the ordered product of the other three links, each matrix element of U_ℓ in the defining representation is an eigenfunction of Δ_ℓ with eigenvalue $-C_2(R_{\mathrm{f}})$, by Peter–Weyl. Hence $\Delta_\ell \,\mathrm{tr}\, U_p = -C_2 \,\mathrm{tr}\, U_p$, and summing over the four links of p, $\Delta\,\mathrm{Re}\,\mathrm{tr}\, U_p = -4C_2\,\mathrm{Re}\,\mathrm{tr}\, U_p$. Summing over plaquettes, $\Delta V = \Delta\sum_p (N - \mathrm{Re}\,\mathrm{tr}\, U_p) = 4C_2 \sum_p \mathrm{Re}\,\mathrm{tr}\, U_p = 4C_2(N|\mathcal{P}| - V)$. □

The gradient estimate

Gradient estimate. On the periodic three-dimensional spatial cubic lattice with four plaquettes per link, for $SU(2)$, $|\nabla V|^2 \le 16\,V$ pointwise on $\mathcal{M}$. For general $SU(N)$, $|\nabla V|^2 \le 16\,N\,V$.

Proof. For one link of one plaquette, the completeness relation of the upper-bound note Corollary 2 gives

$$\sum_a \left|X_a^\ell \operatorname{tr} U_p\right|^2 = \tfrac{1}{2}\Big[\operatorname{tr}(MM^\dagger) - \tfrac{1}{N}\left|\operatorname{tr} M\right|^2\Big] = \tfrac{1}{2}\Big[N - \tfrac{1}{N}\left|\operatorname{tr} U_p\right|^2\Big].$$

For $SU(2)$, writing $\operatorname{tr} U_p = 2\cos\theta$, this is $1-\cos^2\theta = \sin^2\theta = (1-\cos\theta)(1+\cos\theta)$, while $V_p = 2 - 2\cos\theta$, so

$$\left|\nabla_\ell \operatorname{Re}\operatorname{tr} U_p\right|^2 \le \sin^2\theta = \frac{V_p}{2}\Big(2 - \frac{V_p}{2}\Big) \le V_p.$$

Each link lies in four plaquettes, so by Cauchy–Schwarz $|\nabla_\ell V|^2 \le 4\sum_{p\ni\ell} V_p$, and summing over links, with four links per plaquette, $|\nabla V|^2 \le 4\sum_\ell \sum_{p\ni\ell} V_p = 16\sum_p V_p = 16V$. The general case replaces the bound $\sin^2\theta \le V_p$ by $\frac{1}{2}[N - |\operatorname{tr} U_p|^2/N] \le NV_p$, which follows from $|\operatorname{tr} U_p| \ge N - V_p$ when $V_p \le N$ and from the trivial bound otherwise. □

Both identities vanish at $U_p = 1$: a configuration with no magnetic energy has no magnetic gradient, as it must.

The ground-state sum rule

Ground-state sum rule. With $W = V - N|\mathcal{P}|$, $K = \int |\nabla\Omega|^2 d\mu$, $\langle\cdot\rangle = \langle\Omega, \cdot\,\Omega\rangle$ and $\operatorname{Var}_\Omega(V) = \langle V^2\rangle - \langle V\rangle^2$,

$$A'\int W\left|\nabla\Omega\right|^2 d\mu + A'(2C_2 - K)\langle W\rangle + B'\operatorname{Var}_\Omega(V) = 0.$$

Proof. Multiply $-A'\Delta\Omega + (B'V - e_0)\Omega = 0$ by $W\Omega$ and integrate. Since

$$\int W\Omega\,\Delta\Omega = -\int W|\nabla\Omega|^2 - \int \Omega\,\nabla W\cdot\nabla\Omega,$$

$$\int \Omega\,\nabla W\cdot\nabla\Omega = -\tfrac{1}{2}\int (\Delta W)\Omega^2 = 2C_2\langle W\rangle,$$

by the Laplacian identity, so $\int W\Omega\Delta\Omega = -\int W|\nabla\Omega|^2 - 2C_2\langle W\rangle$ and the equation becomes $A'\int W|\nabla\Omega|^2 + 2A'C_2\langle W\rangle + B'\langle VW\rangle - e_0\langle W\rangle = 0$. Now $e_0 = A'K + B'\langle V\rangle = A'K + B'(\langle W\rangle + N|\mathcal{P}|)$ and $\langle VW\rangle = \langle W^2\rangle + N|\mathcal{P}|\langle W\rangle$, so the last two terms combine into $-A'K\langle W\rangle - B'\langle W\rangle^2 + B'\langle W^2\rangle$. □

The sum rule ties the magnetic variance of the ground state to its kinetic energy and to the correlation between magnetic energy and kinetic density. At $g \to \infty$ every term vanishes with B', and at finite coupling it is one exact constraint on the ground-state measure.

SU(3) specialization of the identities

At $N = 3$, $C_2(R_{\mathrm{f}}) = 4/3$, so the actual identities above give

$$\Delta V = \frac{16}{3}(3|\mathcal{P}| - V), \qquad |\nabla V|^2 \le 48V.$$

The sum rule uses $2C_2 = 8/3$. The gradient constant is the stated nonoptimal estimate with four plaquettes per link; it is not a continuum-field supremum bound.

Weighted estimates and global magnetic tails

Weighted identity. For every real Lipschitz weight ρ,

$$A' \int_{\mathcal{M}} |\nabla(e^{\rho}\Omega)|^2 d\mu + \int_{\mathcal{M}} (B'V - e_0 - A'|\nabla\rho|^2) e^{2\rho}\Omega^2 d\mu = 0.$$

Proof. Multiply the eigenvalue equation by $e^{2\rho}\Omega$ and integrate by parts on the closed manifold. Almost everywhere,

$$|\nabla(e^{\rho}\Omega)|^2 = \nabla\Omega \cdot \nabla(e^{2\rho}\Omega) + e^{2\rho}\Omega^2|\nabla\rho|^2.$$

Substitution gives the identity; Lipschitz weights are valid weak test functions on this compact manifold. □

Weighted-mass bound. Put $q = B'V - e_0$, $\Sigma_{\varepsilon_E} = \{q \geq \varepsilon_E\}$, $\varepsilon_E > 0$. If ρ is Lipschitz,

$$A'|\nabla\rho|^2 \leq q - \varepsilon_E/2 \quad \text{on } \Sigma_{\varepsilon_E}, \qquad \rho \leq R \quad \text{on } \mathcal{M} \setminus \Sigma_{\varepsilon_E},$$

then

$$\frac{\varepsilon_E}{2} \int_{\Sigma_{\varepsilon_E}} e^{2\rho}\Omega^2 d\mu \leq K_\rho e^{2R} \int_{\mathcal{M}\setminus\Sigma_{\varepsilon_E}} \Omega^2 d\mu, \qquad K_\rho = e_0 + A'\|\nabla\rho\|_\infty^2.$$

More generally, replacing $\varepsilon_E/2$ in the admissibility inequality by any $\eta > 0$ replaces the left coefficient by η. Thus

$$\int_{\mathcal{M}} e^{2\rho}\Omega^2 d\mu \leq e^{2R}\left(1 + \frac{K_\rho}{\eta}\right).$$

Proof. Drop the nonnegative gradient integral in the weighted identity. On the forbidden set the remaining coefficient is at least η; on its complement it is at least $-K_\rho$ because $V \geq 0$. Split the integral and use normalization. The complementary weighted mass is at most e^{2R}. □

The Agmon distance to $\mathcal{A} = \{q \leq 0\}$ is

$$d(U) = \inf_{\gamma:\, U\to\mathcal{A}} \int_\gamma \sqrt{q_+/A'}\, |d\gamma|.$$

The positive part is essential: the metric vanishes in the allowed region. This distance is Lipschitz, vanishes on $\mathcal{A}$, and satisfies $A'|\nabla d|^2 \leq q_+$ almost everywhere. Indeed appending a short segment bounds its increment by the local weighted length; the bounded weight also bounds its ordinary Lipschitz constant. Among weights vanishing on $\mathcal{A}$ with this gradient constraint, d is maximal, by integrating the constraint along every path. This zero-set and gradient constraint is distinct from the weighted-mass bound's strict margin condition.

For $\rho = (1-\delta)d$, define $\theta_\delta = 1 - (1-\delta)^2 = 2\delta - \delta^2 > 0$. On Σ_{ε_E},

$$q - A'|\nabla\rho|^2 \geq \theta_\delta q \geq \theta_\delta \varepsilon_E.$$

The $\varepsilon_E/2$ margin is therefore guaranteed by this argument only if $\delta \geq 1 - 1/\sqrt{2}$. For arbitrary $0 < \delta < 1$, use the generalized margin $\eta = \theta_\delta\varepsilon_E$ instead. Let $q_{\max} = \max_{\mathcal{M}} q_+$ and $R_{\varepsilon_E,\delta} = (1-\delta)\sup_{\{q<\varepsilon_E\}} d$. Then

$$\int e^{2(1-\delta)d}\Omega^2 d\mu \leq e^{2R_{\varepsilon_E,\delta}}\left(1 + \frac{e_0 + (1-\delta)^2 q_{\max}}{\theta_\delta\varepsilon_E}\right).$$

All constants are finite at fixed lattice and coupling. In particular,

$$R_{\varepsilon_E,\delta} \leq (1-\delta)\,\mathrm{diam}(\mathcal{M})\sqrt{q_{\max}/A'},$$

since $\mathcal{A}$ is nonempty ($V = 0$ is attainable and $e_0 \geq 0$). This is not a uniform constant in volume or coupling. No pointwise bound on Ω follows just by dropping the gradient term in this identity.

An all-path lower bound and the global tail

The constant trial function gives $e_0 \le B'\langle V\rangle_{\mathrm{Haar}} = B'N|\mathcal{P}|$, since each plaquette has a Haar-distributed holonomy. Also $V \le 2N|\mathcal{P}|$, so $q_{\max} \le 2B'N|\mathcal{P}|$.

Global-tail proposition. For any $M \ge \|\nabla V\|_\infty$, $M > 0$, set

$$f = V - \bar{v}, \qquad F(U) = \frac{4}{3g^2M} f(U)_+^{3/2}.$$

Then $d \ge F$. For any $\varepsilon_E > 0$ and $0 < \delta < 1$ set

$$R_F = (1-\delta)\frac{4}{3g^2M}\left(\frac{\varepsilon_E}{B'}\right)^{3/2}, \qquad C_{\varepsilon_E,\delta} = e^{2R_F}\left(1 + \frac{e_0 + (1-\delta)^2 q_{\max}}{\theta_\delta \varepsilon_E}\right).$$

For $\mathbb{P}_\Omega(B) = \int_B \Omega^2 d\mu$,

$$\mathbb{P}_\Omega(f \ge s) \le \min\left\{1,\ C_{\varepsilon_E,\delta}\exp\left[-\frac{8(1-\delta)}{3g^2M}s^{3/2}\right]\right\}, \qquad s > 0.$$

Proof. On any path from U with $f(U) > 0$ to $\{f \le 0\}$, stop at its first zero of f. With arclength parameter r, $|f'(r)| \le M$. The total variation of $\frac{2}{3}f^{3/2}$ along this segment is at least its endpoint difference, even if f is not monotone. Hence

$$\int_\gamma \sqrt{(B'/A')f_+}\, dr \ge \frac{\sqrt{B'/A'}}{M}\int \sqrt{f}\,|f'|\, dr \ge \frac{2\sqrt{B'/A'}}{3M} f(U)^{3/2} = F(U).$$

Taking the infimum proves the all-path lower bound. For an explicit prefactor, apply the weighted-mass bound directly to $\rho = (1-\delta)F$, rather than estimating d on the complementary set. Differentiation gives

$$A'|\nabla F|^2 = \frac{B' f_+}{M^2}|\nabla V|^2 \le q_+.$$

Thus the margin on Σ_{ε_E} is $\theta_\delta\varepsilon_E$, $\rho \le R_F$ on its complement, and $K_\rho \le e_0 + (1-\delta)^2 q_{\max}$. The full weighted mass is bounded by $C_{\varepsilon_E,\delta}$. On $\{f \ge s\}$ the weight is at least $\exp[8(1-\delta)s^{3/2}/(3g^2M)]$, proving the tail. □

For the periodic three-dimensional spatial cubic lattice each link is in four plaquettes. The completeness relation yields $|\nabla_\ell \operatorname{Re}\operatorname{tr} U_p| \le \sqrt{N/2}$ (the upper-bound note, the weighted-mass bound, proof passage). The triangle inequality and sum of squared link components give

$$\|\nabla V\|_\infty \le 4\sqrt{N/2}\sqrt{|\mathcal{E}|} = 2\sqrt{2N|\mathcal{E}|} =: M.$$

In other spatial dimensions or geometries the plaquette incidence must be changed; this constant is specifically for three spatial dimensions. With $P = |\mathcal{P}| = |\mathcal{E}|$ and $s = wP$, the global-tail proposition becomes

$$\mathbb{P}_\Omega(f \ge wP) \le \min\left\{1,\ C_{\varepsilon_E,\delta}\exp\left[-\frac{2\sqrt{2}(1-\delta)}{3}\frac{w^{3/2}}{g^2\sqrt{N}}P\right]\right\}.$$

At fixed $g, N, \varepsilon_E, \delta$, $R_F = O(P^{-1/2})$ and the displayed prefactor is at most $O(P)$. Thus this is a genuine global extensive-tail bound for fixed $w > 0$. When w is a fixed positive multiple of N, the exponent per plaquette is of order N/g^2. Coupling dependence of the prefactor must still be retained in any weak-coupling limit.

For a fixed *total* excess $s = nv$ the exponent instead scales as

$$(nv)^{3/2}/(g^2\sqrt{NP}),$$

which vanishes as $P \to \infty$. A local event need not imply even this total excess, since the complementary plaquettes can lower V. The argument supplies no volume-uniform local estimate.

Exponential comparison and its global threshold

Try $\varphi = e^{-\lambda V/2}$, $\lambda > 0$, as a comparison function for $L = -A'\Delta + (B'V - e_0)$. Using $\Delta\varphi = \varphi[\frac{\lambda^2}{4}|\nabla V|^2 - \frac{\lambda}{2}\Delta V]$ and the Laplacian and gradient estimates,

$$\frac{L\varphi}{\varphi} = B'V - e_0 - A'\frac{\lambda^2}{4}|\nabla V|^2 + A'\frac{\lambda}{2}\Delta V \;\geq\; \Big[B' - 4A'\lambda^2 - 2A'\lambda C_2\Big]V + \Big[2A'\lambda C_2 N|\mathcal{P}| - e_0\Big],$$

for $SU(2)$, using $|\nabla V|^2 \leq 16V$ and the Laplacian identity.

Exponential-comparison lemma. If λ satisfies $B' > 4A'\lambda^2 + 2A'\lambda C_2$, then $L\varphi \geq 0$ on $\{V \geq V_*\}$ with

$$V_* = \frac{(e_0 - 2A'\lambda C_2 N|\mathcal{P}|)_+}{B' - 4A'\lambda^2 - 2A'\lambda C_2},$$

and the maximum principle gives

$$\Omega(U) \;\leq\; \Big(\max_{\{V\leq V_*\}} \frac{\Omega}{\varphi}\Big)\, e^{-\lambda V(U)/2} \qquad \text{for all } U.$$

Proof. Set $u = \Omega/\varphi$. From $L\Omega = 0$, $0 = -A'\varphi\Delta u - 2A'\nabla u\cdot\nabla\varphi + u\,L\varphi$. At an interior maximum of u one has $\nabla u = 0$ and $\Delta u \leq 0$, so the first two terms are ≥ 0 and $u\,L\varphi \leq 0$; since $u > 0$, the maximum lies in $\{L\varphi \leq 0\} \subset \{V \leq V_*\}$. □

Where it stops. With $e_0 \leq B'N|\mathcal{P}|$, the threshold obeys

$$V_* \;\leq\; N|\mathcal{P}|\,\frac{B' - 2A'\lambda C_2}{B' - 4A'\lambda^2 - 2A'\lambda C_2} \;\xrightarrow[\lambda\to 0]{}\; N|\mathcal{P}|,$$

This is an upper bound on the threshold, not a proof that its actual value is asymptotic to the Haar mean. The resulting comparison uses a constant determined on the global sublevel $\{V \leq V_*\}$. A local event need not force a global excess. No volume-uniform local probability estimate follows from the available bounds on that threshold and normalization constant, as in the global-tail proposition above.

Excited states and finite spectral subspaces

For a normalized excited eigenfunction ψ_j with dimensionless energy e_j, positivity is unavailable: real excited eigenfunctions orthogonal to Ω change sign. Nevertheless the identity holds as

$$A'\int |\nabla(e^{\rho}\psi_j)|^2 d\mu + \int (B'V - e_j - A'|\nabla\rho|^2)e^{2\rho}|\psi_j|^2 d\mu = 0.$$

For real eigenfunctions the proof is unchanged; for complex ones test with $e^{2\rho}\overline{\psi_j}$ and take the real part. The weighted-mass bound then applies with e_j and $|\psi_j|^2$, with its corresponding global threshold and constants. It does not require taking $\log\psi_j$ or assuming a positive excited state.

A superposition of eigenfunctions is not itself an eigenfunction. For a whole finite-volume spectral subspace below e_* one can choose a common admissible weight relative to $B'V - e_*$ and bound its weighted norm by summing the individual estimates: if its rank is m and each weighted eigenfunction norm squared is at most C_*, pointwise Cauchy–Schwarz gives a bound mC_* for any normalized superposition. Compactness gives finite rank at fixed e_*. Neither the rank nor the constants have been controlled uniformly in volume or cutoff here. Merely replacing e_0 by the first excited energy e_1 does not prove a uniform low-energy-subspace estimate.

Slice identities and the temporal limit

Fix $G = SU(N)$, $g > 0$, spatial spacing $a > 0$, and a finite periodic spatial box. Let $d\mu$ be normalized product Haar measure on its spatial links. Write a_t for the temporal spacing **in length units** (the physical time step is a_t/c), and $\eta_t = a_t/a$. In the convention of the obligations map, §1 (passage read), the anisotropic Wilson action is

$$\frac{S_E}{\hbar} = \frac{2}{g^2}\left[\eta_t^{-1} \sum_{P \text{ temporal}} (N - \operatorname{Re}\operatorname{tr} U_P) + \eta_t \sum_{P \text{ spatial}} (N - \operatorname{Re}\operatorname{tr} U_P)\right].$$

The Hamiltonian target is H_{KS} in the common setting above. The transfer-matrix references are Lüscher, *Commun. Math. Phys.* **54** (1977) 283–292, and Creutz, *Phys. Rev. D* **15** (1977) 1128–1136 (both metadata level; not used here as a checked proof of temporal-limit uniformity). A formal transfer-kernel expansion is not a ground-state-projector convergence theorem. The temporal length step requires $a_t H/(\hbar c)$ in the exponent, together with a specified topology and spectral control.

Perron-vector slice identity. Fix $a_t > 0$. Assume the physical transfer matrix $\mathcal{T}_{a_t}$ is compact, self-adjoint, nonnegative, and has a simple largest eigenvalue $\lambda_{a_t} > 0$ with normalized positive eigenvector Ω_{a_t}. These are the transfer-matrix hypotheses used below; they are distinct from T1 for H_{KS}. For bounded multiplication F and any boundary vector χ with $\langle \Omega_{a_t}, \chi\rangle \neq 0$,

$$\langle \Omega_{a_t}, F\Omega_{a_t}\rangle = \lim_{n\to\infty} \frac{\langle \mathcal{T}_{a_t}^n \chi, F\mathcal{T}_{a_t}^n \chi\rangle}{\|\mathcal{T}_{a_t}^n \chi\|^2}.$$

For nonnegative χ and the Wilson transfer-kernel representation, the ratio is the expectation in the anisotropic Euclidean slab of $2n$ steps, with boundary weights χ and F inserted at the middle slice. Thus its limit is the marginal $|\Omega_{a_t}|^2 d\mu$. For a general signed or complex χ the spectral formula still holds, but its boundary functional need not be a positive Euclidean probability measure.

Proof. Let P_{a_t} project onto Ω_{a_t}. Compactness and simplicity give

$$r_{a_t} = \|\mathcal{T}_{a_t}(1 - P_{a_t})\|/\lambda_{a_t} < 1, \qquad \lambda_{a_t}^{-n}\mathcal{T}_{a_t}^n \chi = P_{a_t}\chi + O(r_{a_t}^n \|\chi\|)$$

in Hilbert norm. Since $P_{a_t}\chi \neq 0$, the normalized ratio converges to the displayed expectation. Integrating successive transfer kernels is the slab functional by its construction. □

Hamiltonian identity and conditional Wilson limit. The finite-lattice H_{KS} has a unique positive ground state Ω and a gap by T1 of the obligations map, §2 (passage read). Spectral projection therefore gives, directly,

$$\langle \Omega, F\Omega\rangle = \lim_{\tau\to\infty} \frac{\langle e^{-\tau H_{\mathrm{KS}}/\hbar}\chi, Fe^{-\tau H_{\mathrm{KS}}/\hbar}\chi\rangle}{\|e^{-\tau H_{\mathrm{KS}}/\hbar}\chi\|^2},$$

where τ is physical time and $\langle \Omega, \chi\rangle \neq 0$. This is an exact semigroup identity, without invoking a Wilson limit.

To identify it with the preceding anisotropic Wilson marginal, a sufficient additional hypothesis is

$$\|P_{a_t} - P_0\| \longrightarrow 0 \quad (a_t \downarrow 0), \qquad P_0 = |\Omega\rangle\langle\Omega|.$$

Under this hypothesis the positive normalized vectors converge in $L^2(d\mu)$, and their densities converge in $L^1(d\mu)$, since

$$\big\||\Omega_{a_t}|^2 - |\Omega|^2\big\|_1 \leq 2\|\Omega_{a_t} - \Omega\|_2.$$

Consequently the **iterated** limit, first $n \to \infty$ at fixed a_t, then $a_t \downarrow 0$, equals $\langle \Omega, F\Omega\rangle$ for every fixed bounded F. Projector convergence is not proved here. No interchange with the reverse order, no joint limit, and no estimate uniform in spatial volume or spacing is asserted.

Local probabilities and the missing normalized comparison

At fixed a_t, a bound for a fixed measurable slice event A under the infinite-time anisotropic Wilson measure is exactly a bound under $|\Omega_{a_t}|^2 d\mu$. If the projector convergence above holds and

$$\int_A |\Omega_{a_t}|^2 d\mu \le p \quad (0 < a_t < a_{t,0})$$

with the same p, its limit is at most p under $\Omega^2 d\mu$. Bounds uniform in slab length and a_t can pass through the stated iterated limits. Events or couplings that vary in those limits need their own convergence estimates. An isotropic bound at $a_t = a$ is not a bound for the family $a_t/a \downarrow 0$ at fixed a; a four-dimensional block event is not automatically a slice event.

Action cost does not normalize a probability

For a smooth continuum field, the invariant block quantity $\mathcal{F}_B = (\ell^{-4} \int_B |F|^2)^{1/2} \ge \eta/\ell^2$ forces $S_E/\hbar \ge \eta^2/(4g^2)$ in the normalization $(4g^2)^{-1} \int |F|^2$. The action-cost proof owns the inequality, its flow hypotheses and the limitations of sharpness. A Wilson-lattice version additionally needs its field/action comparison.

For a regulated reference law ν and dimensionless action s,

$$\mathbb{P}(A) = \frac{\int_A e^{-s} d\nu}{Z} \le \frac{e^{-q}\nu(A)}{Z} \quad \text{if } s \ge q \text{ on } A.$$

A lower bound on Z relative to the constrained numerator, with boundary, entropy, volume and cutoff control, is still missing. Likewise $e^{-(2/g_E^2)V} = \prod_p e^{-(2/g_E^2)V_p}$ is a local unnormalized weight, not independence of shared-link plaquettes. An increase $\Delta V = nv$ multiplies that weight by e^{-2nv/g_E^2}; it does not identify its local probabilities with the global ground-state tail or equate g_E with g in a limit. Replacing bare g by $g(\ell)$ also needs renormalized measure and observable matching. Thus the target $\mathbb{P}(\mathcal{F}_B \ge \eta/\ell^2) \le Ce^{-c\eta^2/g(\ell)^2}$ remains conditional, not a theorem of the slice identity.

Gaussian benchmarks and kernel tails are different inputs

The specified free-field calculation has three-component variance $g^2/(16\pi^2 t^2)$ and radial logarithmic tail $-24\pi^2\eta_t^2/g^2 + O(\log(\eta_t/g))$ at threshold $|b_t| > \eta_t/t$. Its smearing and component variance determine those constants. They are neither sharp-block nor interacting constants, and a pointwise variance is not a curvature supremum over space and the full flow interval.

The auxiliary kernel estimate uses the budget $A_t = 2\int_0^t \Gamma(s)ds$. A supplied supremum $\Gamma(s) \le \eta/\ell^2$ on that interval, with $\ell^2 = 8t$, gives $A_t \le \eta/4$. A lower block RMS threshold gives no such upper supremum bound. Even a proved probability bound for that block would not license multiplication by $e^{\eta/4}$ on an uncontrolled event, much less a relative Hamiltonian form estimate.

The failed crossover illustrates the distinction. Even assuming unit prefactor and the old target $e^{-c\eta^2/g^2}$, at $\eta = 2g^2/c$ its upper bound is $e^{-4g^2/c} \to 1$, establishing no rarity. With further matching assumptions, multiplying by $e^{\eta/4}$ would give the schematic exponent $-c\eta^2/g^2 + \eta/4$, whose zero is $g^2/(4c)$; this algebra is not a sharp crossover or error theorem. Increasing the truncation range controls the absolute auxiliary tail for finite budget. A signed operator comparison remains a separate requirement even when a bad set is rare.

Consequence for STATE

The exact magnetic identities, weighted/global tail with explicit prefactor, and fixed-spacing slice identity are maintained together. The latter's Hamiltonian limit is conditional. Global energy

control, normalized local rarity, auxiliary kernel locality and signed spectral control have distinct assumptions. No local suppression theorem or continuum physical mass gap is established by combining their names.